Basic Economic Concepts

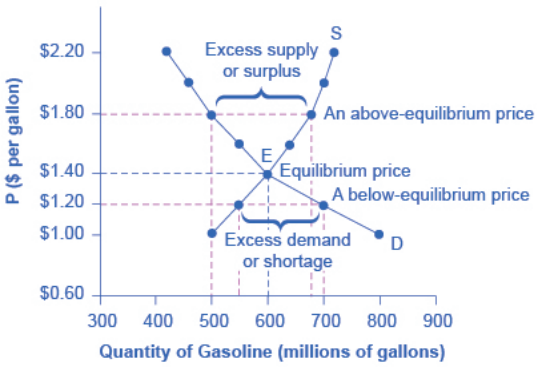
Introduction

- **Terminologies:**
  - **Opportunity Cost** - The cost of any action in terms of what you could have done instead
  - **Constrained Optimization**
    - Constrained -> Every decision has an `opportunity cost`
    - Optimization -> Make yourself as well off as possible
  - **Model**
    - To simplify things down *(strip down the variables e.g. Occam razor)*
    - Intuitively
    - Graphically
    - Mathematically

Supply and Demand

> Most economics model don't represent the reality instead the are trying approximate it

- **Supply and Demand Model**
  - **Demand -** How much people want it! *(The greater the demand, the higher the price)*
  - **Supply -** How much is available *(The greater the supply, the lower the price)*
  - **Equilibrium point -** The point in which both the consumer and the provider are happy to purchase and provides the goods *(This is where the "demand" curve intersects with the "supply" curve)*
    #+name: Equilibrium Point

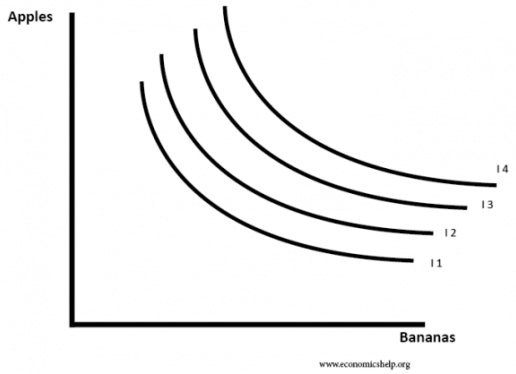


Market Economy vs Command Economy

- **Command Economy:**
  - Trying to even things out among people *e.g. Soviet Union economy*
  - Government decides the economy *(Too much to handle -> shortages, corruption, waiting list, etc)*
- **Market Economy:**
  - Don't try to even things out instead let,
  - Individual decides their own needs and production

Preferences

- **Preferences:**
  - Can be either `indifferent` or `prefer`
  - Logically consistent
  - More is better *(to some extend)*
  - `preference curve` - A condensed or simplify preference of some aspects
- **Properties of a map of indifference curve:**
  1. `consumer` prefer higher indifference curves
  2. Are downward sloping
  3. Never cross
  4. There is **one** indifference curve through each possible `bundle`
     #+name: Indifference Curves



- A level of `total ultility` represents the level of satisfaction when choosing a `bundle` -> *higher level of total utility leads to better satisfaction*

Utility Functions and Marginal Utility

- **Utility function -** A mathematical expression that translates bundles of goods into a single valuation of utility
- **Marginal Utility -** The change in utility from each additional good you consume

> With each extra unit of a good increase `utility` by less than the previous unit of the good this is called **Diminishing Margin Utility** *(DMU)*

> Different goods come with different level of `DMU` *e.g. socks vs newspaper*

Application - Soda pricing

> One of the best examples of diminishing marginal utility is how stores price different size of goods. It shows how `utility` and `supply-and-demand` works together

- Demand is high from **some** drink, but NOT so high for **additional** drink
  - On a hot day demand for a drinks is higher but demand for **additional** drink will be lower *-> Diminishing marginal utility in action*
  - Lower price could be charge for **additional** drink

The Math Behind Utility Functions

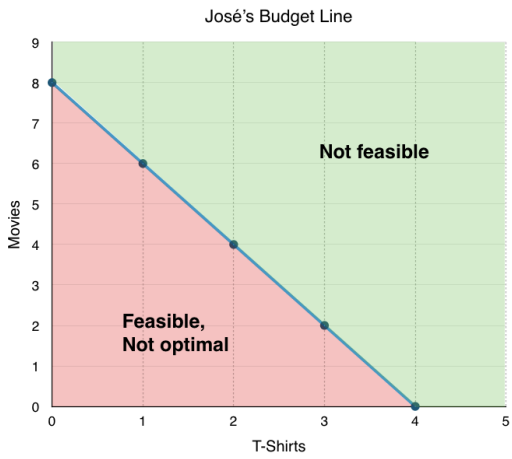
- There are lot of different `utility functions`
- A `utility function` should:
  - Be realistic
    - More is better
    - Have a `diminishing` component
  - Easy to work with
  - e.g. $U = \sqrt{(P * C)}$

Budget Constraints

> A Budget constrain measures how much of the goods an individual can afford to buy. Maximum budget constraints is defined by an individual paycheck

Introduction to Budget Constraints

- **Budget Constraint properties:**
  - Assuming individual **budget constraint** is their **incomes** -> *living from paycheck to paycheck*
  - Form a horizontal line graphically
  - **Opportunity cost** is represented by **price ratio** *(slope of the budget constraint curve)*

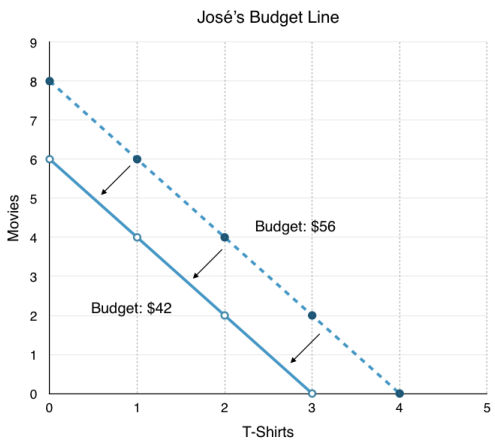


> Opportunity cost can be represented by the value of the `slope` of a curve that is: $m = \frac{\Delta y}{\Delta x}$

Budget Constraints with Changing Income and Prices

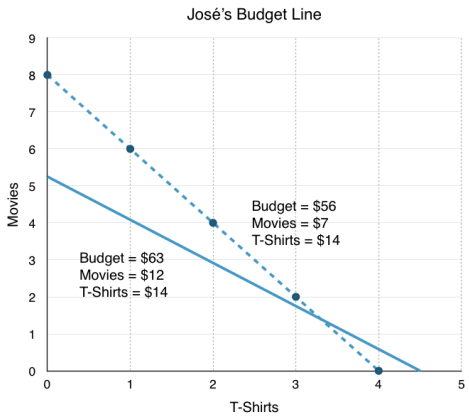
> Is it possible to manipulate the budget constraints to represent a change in price or a change in income

- **Change in income**
  - `slope` stays the same
  - new `budget curve` is parallel to original curve
    #+name: Changes in budget



- **Change in price**
  - `slope` is altered
  - new `budget curve` can shift inwards/outwards
    - Shifting inwards -> more affordable
    - Shifting outwards -> less affordable

    #+name: changes in price



Utility Maximization

> Conceptually model the way in which individuals actually make decision
>
> 1. Concept of maximizing utility
> 2. Example of intuition
> 3. Graphical approach
> 4. Real-world example

> 5. Numerical examples *(optional)*

> Economics is about constrained optimization. It tries to optimizing utility subject to constraint imposed by budgets

Marginal Analysis

> **Marginal Analysis -** A set of tools use to approximate `utility maximization` methodologically

- **Marginal Benefit** *(MB)*
  - Increase in utility from one more unit
  - Can be express as $\frac{\text{marginal utility of pizza}}{\text{marginal utility of cookie}}$
- **Marginal Cost** *(MC)*
  - Opportunity cost of next unit
  - `MC` = `opportunity cost` = `slope` = $\frac{\Delta \text{pizza}}{\Delta \text{cookie}}$
- Comparing `MB` and `MC` of the **next** unit
  - if `MB` > `MC` -> Consume more goods *(climbing up the hill)*
  - if `MB` < `MC` -> Consume less goods *(climbing down the hill)*
  - if `MB` = `MC` -> Stop *(on the top of the hill)*

Graphical Interpretation